

MediPilot: Asymmetric-Autonomy AI for Continuous Re-Triage in the Indian ED



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ABSTRACT

Emergency department (ED) crowding kills through a structural defect, not through individual error: acuity is assigned once, at arrival, and the queue that results is static while the patient’s physiology is not. Emergency Severity Index mistriage affects 32.2% of US ED visits, and India had no nationally standardised ED triage protocol before the AIIMS Triage Protocol (ATP). We present *MediPilot*, a continuous re-triage platform built on a single architectural invariant: **the system may raise a waiting patient’s priority autonomously and may never lower it below a human-assigned level**. Escalation costs one nurse assessment; de-escalation can cost a life, so it stays with a human. We show how this invariant propagates into every layer — outcome-derived rather than nurse-imitating labels, leakage control enforced mechanically in CI after the Epic Sepsis Model’s failure (AUC 0.63, sensitivity 33%), an LLM restricted by output grammar to perception, conformal abstention rather than silent guessing, and a five-minute re-scoring cycle that is both the primary safety mechanism and the commercial wedge. We specify the data acquisition path, training methodology, evaluation protocol (under-triage at fixed over-triage; equalised false-negative rate; stepped-wedge causal attribution), tiered edge deployment at a target of under Rs. 5 per encounter, and the CDSCO/ICMR/DPDP compliance route — including the gap where ABDM consent does not by itself satisfy DPDP Act 2023 for model-input use.

Keywords: clinical decision support • emergency triage • asymmetric cost learning • conformal prediction • federated learning • SaMD regulation • Indian healthcare

1 INTRODUCTION

The danger of an ED waiting room is not that patients arrive critically ill; it is that they *become* critically ill after being assigned a low priority, and no mechanism exists to notice. A patient with mild chest discomfort receives a Yellow designation and sits down. Twenty minutes later they are hemodynamically unstable. The triage nurse is by then registering three new arrivals. The patient deteriorates unseen.

This is the **single-point assessment model**, and it is a property of the process rather than of any clinician within it. Its failure rate is measurable: **32.2%** of total US ED visits are affected by ESI mistriage [9]. In India the position was worse until recently — no evidence-based national triage standard existed before the AIIMS Triage Protocol, a three-tier Red/Yellow/Green system since adopted by AIIMS Bhubaneswar, CMC Vellore, GTB Delhi and the Kerala DHS [3]. Most Indian EDs are staffed by general practitioners or residents without formal emergency-medicine training [2], so any workable system must run in a district hospital with one physician and one nurse as reliably as in a tertiary centre.

Contributions. (i) A decision-theoretic autonomy boundary — autonomous escalation, never autonomous de-escalation — and its propagation as a hard constraint through six architectural layers (§3–§4). (ii) A label and feature design that avoids both nurse-label imitation and the target-leakage failure that broke the Epic Sepsis Model, with leakage enforced by CI rather than by policy (§2, §5). (iii) An evaluation and rollout protocol whose primary metric is under-triage at fixed over-triage, with fairness defined as equalised false-negative rate (§7). (iv) A deployment and governance path specific to Indian infrastructure and regulation (§8).

2 WHY PRIOR CLINICAL AI FAILED

Wong et al. [12] externally validated the Epic Sepsis Model across 38,455 hospitalisations at the University of Michigan (Dec 2018–Oct 2019). The model was already deployed at

hundreds of US hospitals.

ESM (external)	AUC	Sens.	Spec.	PPV
Reported performance	0.63	33%	83%	12%

Two-thirds of sepsis cases missed; 109 alerts raised per true positive found.

The root cause was **target leakage**: the model used antibiotic-order timing as an input feature. It therefore detected *clinician suspicion* rather than the underlying physiology — it learned to imitate the process it was meant to augment, and failed exactly where that process was already failing.

Structural response. MediPilot maintains a **temporal feature registry** recording, per feature, its earliest availability time, source system and leakage class (SAFE / CONDITIONAL / PROHIBITED). Any feature downstream of clinician suspicion is PROHIBITED, and an **automated leakage lint** in CI fails the build if one appears in a training feature set. Inclusion is mechanically impossible, not procedurally discouraged (§5).

A second precedent constrains the language model. Haim et al. [6] found GPT-4 systematically miscalibrated against trained clinical raters, with a documented tendency to *over-estimate* severity. MediPilot therefore gives the LLM no production rule for acuity at all (§4).

3 THE ASYMMETRIC AUTONOMY PRINCIPLE

MediPilot may escalate a waiting patient’s queue position autonomously. It may never de-escalate a patient below a human-assigned level autonomously.

This is the correct answer to an asymmetric cost structure, not a policy preference. An unwarranted escalation (Yellow→Red on a stable patient) costs one brief nurse assessment and causes no patient harm. An unwarranted de-escalation (Red→Yellow on an evolving STEMI) costs death or permanent disability.

Let p be the calibrated probability of a 24-hour critical event and let $R = c_{FN}/c_{FP}$ be the site’s miss-to-false-alarm

cost ratio. Escalation is optimal when $c_{FN} p \geq c_{FP}(1 - p)$, giving the operating threshold

$$p^* = \frac{c_{FP}}{c_{FP} + c_{FN}} = \frac{1}{1 + R}.$$

With $R = 500$ (tertiary centre), $p^* \approx 0.002$; with $R = 100$ (district hospital), $p^* \approx 0.010$. R is a **per-hospital configurable parameter** set by clinical leadership at deployment and reviewed at each federated update cycle — it is not a global constant chosen by the vendor.

The same ratio enters training as an asymmetric loss (§6), so the threshold and the objective agree rather than being reconciled after the fact.

4 SYSTEM ARCHITECTURE

MediPilot is a six-layer pipeline (Fig. 1) closed by two feedback loops.

4.1 L0 — Capture and the feature registry

Data arrives in three windows with sharply different availability (Fig. 2). Every field is registered with its earliest availability time and leakage class before it can be used. **Operational context** — census, boarding count, staff on shift, bed and ventilator availability — is ingested by the routing layer L4 *only*, and is explicitly excluded from the risk layers L2–L3, so physiologic risk is never a function of how busy the department is.

4.2 L1 — The LLM structurer, perception only

Audio $\xrightarrow{\text{IndicWhisper INT8 ASR}}$ tokens $\xrightarrow{\text{grammar-constrained decoding}}$ SNOMED-mapped schema fields. The output grammar **contains no production rule** for a free-text diagnosis or an acuity recommendation, so the failure mode reported by Haim et al. cannot be expressed. The nurse speaks normally; the system fills the form and reads the vitals back for correction before committing.

4.3 L2 — Risk backbone, late fusion, three heads

- **GBDT** (XGBoost/LightGBM) over tabular T0–T1 fields: 100% of patients, millisecond latency, native missing-value handling.
- **GRU / small transformer** over irregular serial vitals — this is the *deterioration-while-waiting* signal, invisible to any snapshot model.
- **Multilingual IndicBERT** over chief complaint and nurse narrative.

For ABHA-linked patients a self-supervised next-event transformer (ETHOS/CLMBR-style) adds a history embedding. **Train-time modality dropout** forces usable estimates from any subset, because missing data is the default case in an Indian ED, not the exception.

4.4 L3 — Calibration, uncertainty, abstention

An overconfident wrong prediction is more dangerous than an uncertain correct one. **Per-site isotonic/Platt calibration** makes outputs interpretable as local frequencies. **Split conformal prediction** ($\alpha = 0.10$) returns an *acuity set* with a formal $\geq 90\%$ coverage guarantee instead of a spuriously precise point estimate; **Mondrian conformal prediction** holds that coverage separately within each demographic

and linguistic subgroup. An **OOD abstention gate** routes unfamiliar patients to manual nurse assessment with an explanatory note. Abstention is visible, never silent.

4.5 L4 — Decision and routing

L4 solves a constrained assignment problem: maximise expected clinical value subject to hard rules no optimiser component may override.

- A Red patient is never queued behind a Green patient under any resource state.
- Suspected STEMI routes to the cath lab regardless of throughput impact.
- A normal SpO₂ reading **alone** is never grounds for downward queue movement (see box below).

Bandit/RL components act only on the soft dimension — which of several equally safe rooms — inside a runtime safety filter that rejects protocol-violating actions. **Mass-casualty mode**, which switches the objective from individual expected-value to resource-constrained population benefit, is activated only by an authenticated senior clinician. The system never concludes on its own that a mass-casualty event exists.

Pulse-oximetry bias — a non-overridable rule. Sjöding et al. [11] found occult hypoxemia in **11.7%** of Black versus **3.6%** of White patients (single centre), and **17%** versus **6.2%** in a multicentre VA cohort. A normal SpO₂ therefore carries no de-escalation authority in L4. The ambient pallor module is calibrated per Fitzpatrick stratum with false-negative-rate equalisation across strata.

4.6 L5 — Presentation: one card

The nurse sees a recommended acuity level, the **two strongest supporting factors**, **one opposing factor**, and a confidence band. The opposing factor is a deliberate falsification target: it forces engagement with reasoning rather than with a number, which is the standard defence against automation bias. Accept or override is one touch; an override requires a stated reason. No acuity level is ever spoken aloud within patient earshot, and the hall speaker calls token numbers only.

4.7 Feedback loops

Loop A (5 min). Every waiting patient is re-scored. A rise triggers an escalation alert; a fall triggers nothing autonomous. This loop is the primary safety mechanism of the system — it is what makes the corridor observed. **Loop B (continuous).** Overrides plus eventual outcomes flow into federated training. Raw patient data never leaves the institution.

5 DATA, LABELS, AND ACQUISITION

5.1 Label strategy: outcomes, not nurse labels

With ESI mistriage at 32.2%, roughly one nurse label in three is wrong; training a model to reproduce them would encode the error. MediPilot instead trains on a **composite outcome label** extractable from existing admission, discharge and billing records — no clinician chart-review project is required, which removes the usual blocking dependency on scarce clinician time.

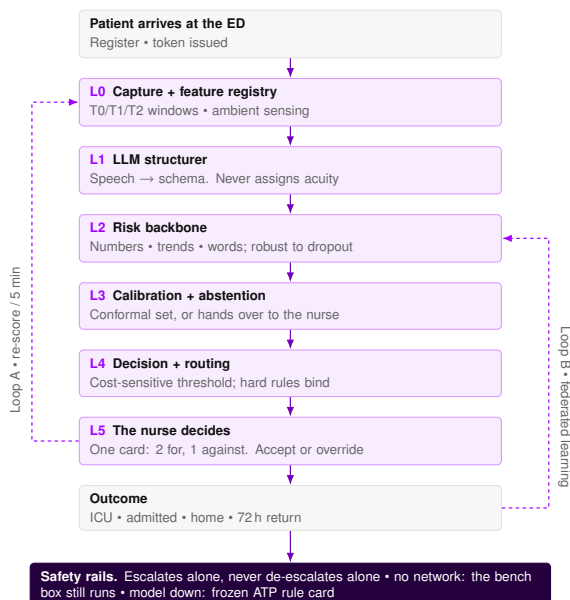


Figure 1. The MediPilot pipeline. Loop A re-scores every waiting patient every five minutes and can only move a patient up; Loop B returns overrides and outcomes to training without moving raw data off site.

Label	Definition
24 h critical composite	Death, ICU admission, intubation, vasopressor start, or emergent OR/cath lab within 24 h
Hospital admission	Any inpatient admission from the ED visit
ED length of stay	Continuous proxy for resource consumption
72 h return visit	Signal of an under-treated first presentation

5.2 Counterfactual censoring

A correctly escalated Red patient who is treated immediately looks “not sick” in raw outcome data — the intervention erased the evidence. Two corrections: **inverse-propensity weighting** on the original triage assignment, and a **physiologic-derangement secondary label** (worst vitals in the first four hours), which is less treatment-contaminated and constrains the primary label through multi-task learning.

5.3 Acquisition in four phases

Phase 0 builds and validates the architecture on MIMIC-IV-ED, eICU and HiRID (~425k ED visits); Xie et al. [13] benchmark LR, RF, GBDT, MLP, Med2Vec and LSTM on this data, with the best models near **0.84 AUROC** on hospitalisation, critical-outcome and 72-hour reattendance tasks. This is a US population and supports no India deployment claim. **Phase 1** takes retrospective exports from 2–3 Indian partner hospitals (150k–300k visits over 3–5 years) to build the India-specific backbone. **Phase 2** is a 3–6 month silent prospective bench capture, which is the only way to obtain real-time signals absent from any retrospective export. **Phase 3** opens the federated network, where each new site contributes weight updates and never raw data.

Cold start is published, not hidden. The system abstains on roughly **30% of patients in month one**, falling below **10% by month six** as local calibration data accumulates. This trajectory is a monitored metric shown to the partner hospital from day one.

6 TRAINING METHODOLOGY

Self-supervised pretraining. Masked modelling over clinical event sequences (vitals, labs, medication administrations) and contrastive learning over vitals trajectories. The resulting backbone serves three purposes: feature extraction for L2, density estimation for L3’s OOD gate, and patient-history encoding for ABHA-linked records.

Phenotype clustering. Unsupervised clustering surfaces rare but deadly subgroups — afebrile elderly sepsis, silent MI, shock presenting as abdominal pain — which are then *explicitly upweighted* in the cost matrix. These are the under-triage classes that kill, and they are too infrequent to be protected by average-case loss.

Cost-sensitive multi-task fine-tuning. Five heads (24 h critical composite, admission, ICU admission, ED length of stay, 72 h return) share the trunk, trained under the asymmetric loss implied by R in §3.

Rare-event augmentation. Critical deterioration occurs in roughly 1–3% of ED presentations. Diffusion models outperform GAN and VAE baselines on both fidelity (Jensen–Shannon divergence, pairwise correlation) and privacy risk (membership inference resistance) for synthetic EHR time series [10]. Synthetic cohorts enter the pipeline only after *train-on-synthetic / test-on-real* validation and a membership-inference privacy audit.

Federated learning. FedProx-style adaptive aggregation handles non-IID site data, with per-site personalisation heads over a shared trunk, secure aggregation and differential-privacy noise — the technical basis for the DPDP posture in §9.

Active learning. The human adjudication panel reviews only the highest-uncertainty cases, ranked by conformal set width and OOD score, which maximises label yield per clinician-hour.

7 EVALUATION AND ROLLOUT

Primary safety metric: under-triage rate at a fixed over-triage rate. Not accuracy, and not AUROC — at 1–3% event prevalence, AUPRC is the informative summary and AUROC flatters every model. Calibration is reported as ECE with reliability slope and intercept; **ECE < 0.05 across all acuity strata and demographic subgroups is a mandatory go-live criterion**, evaluated on shadow-mode data. Decision-curve (net-benefit) analysis is reported across the plausible threshold range rather than at a single operating point.

Fairness is equalised false-negative (under-triage) rate across age, sex, language group, payer type, skin tone and arrival mode — explicitly *not* demographic parity. Demographic parity would require under-triaging genuinely high-risk groups to equalise recommendation rates, which is the precise inverse of the safety objective.

External validation per site. Every new deployment completes external validation before any clinical exposure. This is the direct lesson of Wong et al., where internal validation substantially overstated real performance; shadow mode generates the site-specific validation set.

Rollout. A 3–6 month **shadow mode** logs predictions against outcomes with zero clinical exposure, followed by

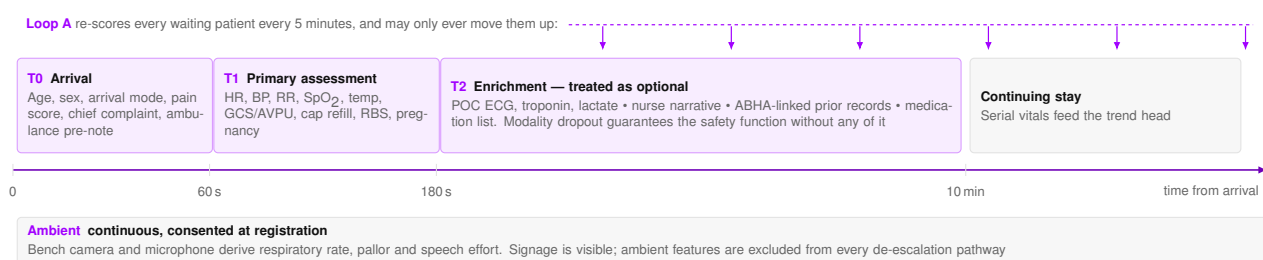


Figure 2. Data availability across the first ten minutes. Each window is a different feature-availability regime, so the model is trained to produce a usable estimate from T0 alone and to treat everything after it as enrichment rather than as a requirement.

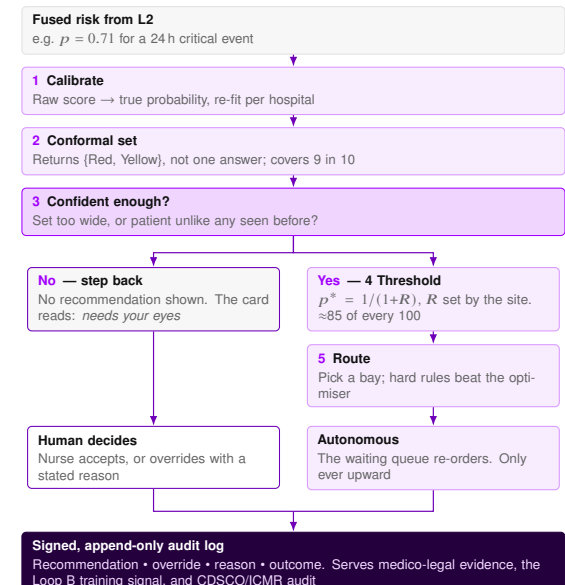


Figure 3. The decision path from a fused risk score to an action. The gate at step 3 is the abstention boundary; the split at the bottom is the autonomy boundary — the queue may re-order itself upward without a human, but every other outcome waits for one.

a **stepped-wedge cluster-randomised** rollout across wards and shifts. Any claim of causal effect on door-to-doctor time, time-to-antibiotic, left-without-being-seen rate or length of stay must rest on that design — an observational pre/post comparison cannot support it, and we will not present one as if it could.

8 DEPLOYMENT

MediPilot must run where the patients are, which includes hospitals with intermittent power and no modern HIS. All inference is local to the bench device; the network is required only for scheduled federated weight synchronisation and optional ABHA enrichment. EHR integration follows the **HL7 CDS Hooks** specification [7] for synchronous, workflow-triggered support, with a standalone tablet interface for sites without an integrable HIS.

	Setting	Hardware	Model path
T1	Metro tertiary / AIIMS-class	On-prem GPU	Full ASR + LLM; full ETHOS/CLMBR foundation model
T2	District, 50–200 beds	Edge GPU (Jetson-class)	Distilled INT8 path; lightweight encoder
T3	PHC / sub-district	CPU-only edge	Rule-based ATP fallback; no foundation model

Unit economics. The target is **under Rs.5 per patient encounter**, achieved by a cascade: the GBDT head runs on 100% of patients at millisecond cost, while the LLM and foundation-model paths fire only on the ~15% flagged ambiguous. Marginal cost falls as sites join the federated network, which keeps the system reachable for district and government facilities rather than only for private tertiary chains.

9 GOVERNANCE AND COMPLIANCE

MediPilot is Software as a Medical Device under CDSCO, **Class B or C**; we design to Class C throughout as the conservative reading. CDSCO’s draft **Algorithm Change Protocol** (21 Oct 2025) [4] — India’s analogue to the FDA/Health Canada/MHRA PCCP of Aug 2025 [5] — provides the route for retraining without full resubmission, and the federated update cycle (§6) is designed to fall inside it. Data governance, algorithmic transparency, human oversight and independent audit follow ICMR’s March 2023 ethical guidelines [8].

The consent gap. ABHA consent artifacts — FHIR-based exchange across **46.8 crore** ABHA IDs and 31 crore linked records as of Oct 2023 [1] — **do not by default satisfy DPDP Act 2023 §6(1)**. ABDM consent covers *record access*; it does not cover the use of those records as *AI model input features*. MediPilot therefore issues a separate, purpose-specific DPDP consent artifact per processing purpose. Ambient monitoring carries its own consent, taken at registration and posted on visible signage.

Every recommendation, every override with its stated reason, and every eventual outcome is written to a cryptographically signed, append-only log serving three consumers: medico-legal evidence, the Loop B training signal, and regulatory audit.

9.1 Leakage lint — the enforcement artifact

The registry claim of §2 is only credible if a build can fail because of it. Each feature carries a registry entry, and CI rejects any training config whose input set intersects the prohibited class:

```
feature: antibiotic_order_to_first_dose
downstream_of_suspicion: true
leakage_class: PROHIBITED
permitted_use: [outcome_label, eval_endpoint]
$ mediopilot-lint train.yaml
FAIL 1 PROHIBITED feature in input_features
```


10 FAILURE MODES

Failure	Mitigation
Missing data as the norm	Train-time modality dropout; the T0-only configuration is an explicitly tested worst case
Atypical pre-senters	Phenotype clustering plus cost-matrix up-weighting, treated as the deadliest under-triage class
Power / network loss	Offline-first edge inference, UPS operation, store-and-forward sync
Model service outage	Hard fallback to the frozen AIIMS-ATP red-flag rule card [3]. The UI never shows a blank state
Pulse-oximeter bias	Normal SpO ₂ carries no de-escalation authority; Fitzpatrick-stratified pallor calibration
Adversarial gaming	Complaint-text distribution tracked against outcome correlation as a drift signal, reviewed by clinical governance
Linguistic diversity	Code-mixed IndicWhisper ASR, multilingual IndicBERT, per-language under-triage rate as a mandatory fairness metric

11 ANTI-DECISIONS, IMPACT, AND CONCLUSION

What MediPilot must not become. No virtual-nurse persona or avatar, which manufactures false clinical authority. No spoken acuity level within patient earshot, for dignity and panic reasons in shared waiting halls. No LLM-assigned severity, which is structurally prohibited in L1 rather than merely discouraged. No autonomous de-escalation, ever. And no chart-review labelling project — outcome-derived labels route around it entirely.

Who this helps. The single triage nurse under load gains a continuous monitoring layer she cannot physically provide. The patient deteriorating in the corridor is re-scored every five minutes rather than every time someone happens to look. The district hospital without a certified emergency physician [2] gains structured decision support grounded in the AIIMS-ATP foundation [3].

Go-to-market wedge. The first shipped product is the *deterioration-while-waiting re-triage module* alone. It carries the highest clinical value and the lowest regulatory exposure, because it never assigns an initial acuity — it only watches a queue a human already ordered. It requires near-zero workflow change, and it generates the site-specific outcome data that every later capability depends on.

Moat. A well-funded competitor can rebuild the technical stack in a year. It cannot rebuild years of site-specific calibration and federated override history across dozens of Indian EDs. Each new hospital improves the model for every other one — a compounding advantage structurally unavailable to a single-site or late entrant.

Conclusion. The waiting-room death is a systems failure with a specific shape: one assessment, then silence. MediPilot's contribution is not a better score but a better contract — a machine that is allowed to raise an alarm on its own and is never allowed to lower one, that abstains out loud when it does not know, and that is evaluated on the patients it misses rather than on the average it gets right.

MediPilot • “Nobody waits unseen.”

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DESIGN COMMITMENTS AT A GLANCE

Commitment	Enforcing mechanism
Never de-escalates alone	Architectural constraint in L4 and Loop A, not a configurable setting
Never learns clinician suspicion	PROHIBITED leakage class; CI build failure (\$9.1)
Never assigns acuity by LLM	No production rule for acuity exists in the L1 output grammar
Never guesses silently	Conformal abstention gate; the card says so on the ~30%→<10% it declines
Never trades safety for throughput	ED load reaches L4 only; it is excluded from L2–L3 risk scoring
Never claims impact it cannot attribute	Stepped-wedge cluster randomisation, not pre/post comparison

Note on format. No style guidelines were issued for this submission at the time of writing; the document follows standard technical white-paper conventions and can be reformatted on request.